

Full Length Article

A DFT study of methanol synthesis from CO₂ hydrogenation on Cu/ZnO catalystXingzi Wang^{a,*}, Hai Zhang^{a,*}, Huang Qin^a, Kunming Wu^a, Kai Wang^a, Junfang Ma^b, Weidong Fan^a^a Institute of Thermal Energy Engineering, School of Mechanical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China^b Weichai Power Co. Ltd, 197A Fushou St. E, Weifang 261016, China

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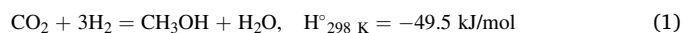
ABSTRACT

The usage of CO₂, a vital carbon source, is of great application value in carbon neutrality. Hydrogenation is one of the most promising approaches to convert the CO₂ into high-value chemicals like methanol. The development of the hydrogenation of CO₂ mainly lies in the design of safe and efficient catalysts. Focusing on the mechanism and the interface effects, the hydrogenation of CO₂ to methanol over Cu/ZnO-based catalyst was investigated in this work. To study the enhancing effect of metal promoters, atomic doping was simulated on Cu/ZnO-X(Al, Mg, Ga, Pt, Pd, Au) catalyst. Based on the designed atomic doped mode, density functional theory (DFT) calculation was conducted to analyze the adsorption of intermediates, thermodynamic reaction path, and kinetics of CO₂ methanolization. The results show that Cu-ZnO heterostructure improves the HCOO path of CO₂ hydrogenation by metal-support interaction (MSI). A linear relationship between the adsorption energy of the intermediates via hydrogenation process was found with the correlation coefficient $R^2 \approx 0.8$. The order of the highest activation barriers for the overall reaction is Cu/ZnO-Au > Cu/ZnO-Mg > Cu/ZnO-Pd > Cu/ZnO-Pt > Cu/ZnO-Ga > Cu/ZnO > Cu/ZnO-Al. The diverse performance of various metal promoters was ranked as Al > Ga > Mg, Pt > Pd > Au. Our simulation work well corresponds the previous experimental results conducted by other scholars and will provide guidance for future design of the high-efficient catalysis for CO₂ hydrogenation.

1. Introduction

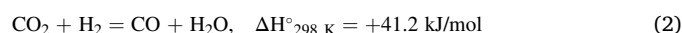
Since the industrial revolution, human activities have released tremendous quantities of CO₂ into the vulnerable atmosphere and have had irreversible negative impacts on global climate. Phrases like ‘zero-carbon’, ‘carbon-negative’, ‘carbon neutrality’ etc. gradually became popular hashtags on global media. Strategies spring up to counteract the unavoidable emission of CO₂ during daily industrial procedure. The concept of ‘Carbon Capture, Usage and Storage’ (CCUS), including the conversion of CO₂ to high-value chemicals, especially methanol, by using renewable resources, stands out as a feasible and promising one [1–3].

At ambient temperature, the hydrogenation of CO₂ to methanol is exothermic:

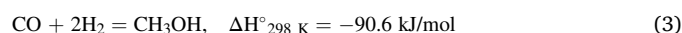


On the contrary, its main competing reaction, the reverse water gas

reaction (RWGS), is an endothermic reaction:



At the same time, CO can also be hydrogenated to methanol:



Among all reactions, since the reaction (1) is exothermic and the number of molecules gradually decreases in the forward reaction direction, therefore high pressure and low temperature are favorable for the forward reaction to generate methanol. The reverse water gas reaction (2) that generates CO is endothermic, and high temperature will promote the generation of CO. As a result, the whole process mainly relies on the design of the efficient catalyst with high performance at high pressure and low temperature that can suppress the reverse water gas reaction, reduce the production of by-product CO, and improve the selectivity of methanol [4].

Industrially, Cu/ZnO/Al₂O₃ catalyst was developed to obtain the

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methanol from synthesis gas ($\text{CO}_2/\text{CO}/\text{H}_2$) at 493–573 K and 5 ~ 10 MPa at the early age [5,6]. And then extensive efforts have been paid on the application of Cu-based catalysts on methanol synthesis from CO_2 hydrogenation. The first studies began from the pure Cu catalyst. But researchers soon found it inadequate to make an efficient conversion [7,8]. Various oxides, ZnO, CeO_2 , TiO_2 , ZrO_2 , SiO_2 etc. have later been proved theoretically and experimentally to help preventing the sintering, expanding the specific surface area of Cu and thus increasing the active sites [9–14]. However, the two-component catalyst such as Cu-ZnO does have drawbacks, including the poor thermal stability, short lifetime and extreme sensitivity to toxic substances such as CO [7–8,14]. Hence, the ternary oxides catalysts were proposed to fix these flaws. Cu/ZnO/ Al_2O_3 was then successfully developed and put into production as one of the first commercial methanol catalysts [15,17]. Li et al. [16] also created a highly selective CuO/ZnO/ ZrO_2 catalyst.

Previously, many studies have focused on the reaction mechanisms of CO_2 hydrogenation over Cu-based catalysts. Traditionally, two different routes of CO_2 hydrogenation over Cu-based catalysts have been proposed (see Fig. 1) [5,8–13]. First, C-determined hydrogenation to HCOO^* species. Second, O-determined hydrogenation to COOH^* . Considering their stability, the formate and the carboxyl are often depicted as a crucial intermediate.

With regards to the mechanism over Cu/ZnO-based catalysts, the formate pathway takes the lead [8,15–21]. Diverse analysis, aiming to thoroughly understand the pathway, figure out the key active site, and thus find the potent promoter, have been performed. Kattel et al. [18] studied ZnO/Cu(111) catalysts and observed a large amount of HCOO^* on the surface by Popper Optical Methods, implying that the formate pathway is preferred. In the meanwhile, they found that ZnO increases the binding ability of the catalyst to O-containing substances and thus accelerating the hydrogenation process. Yang et al. [19] also came to a similar conclusion by validating that adding additives that promote HCOO^* hydrogenation elevates the methanol yield significantly.

In the meanwhile, researchers also tried to dig into the reason why Cu/ZnO is more efficient than Cu and they focused on the interface effect of oxide materials and active metal. Tauster et al. [22] firstly reported an encapsulation phenomenon caused by the migration of TiO_2 carriers to the active metal surface under high temperature reducing condition, which deprive the chemisorption ability of the catalyst. They name it as “strong metal-support interaction (SMSI)” [23]. From then on, the concept of the metal-support interaction (MSI) is often emphasized to study the interface effects [23,24]. Subsequently, it was also reported that both the electron transfer between the metal and the support and the electron enrichment at the interface could have a significant effect on the performance of the catalyst, the phenomenon of which is called electronic metal-support interaction (EMSI) [25].

Moreover, to gain better methanol selectivity via CO_2 hydrogenation, the activation of CO_2 and the adsorption of key intermediates should be enhanced [12,26–31]. Studies suggested that the modification by another metal or their oxides based on Cu/ZnO catalyst help optimize these procedures. For example, Wang et al. [27] doped Cu/ZnO/ ZrO_2 catalysts with metal elements (Al, Fe and Mg) by co-precipitation method and found that the crystal configuration of -ZrO_2 was maintained in all doping cases, while CO_2 conversion and methanol selectivity were both improved. Sloczynski et al. [28] investigated the doping

of Cu/ZnO/ ZrO_2 catalysts with various elemental oxides such as Ga, In, Mg, Mn. The addition of metal oxides was found to improve the conversion efficiency and methanol selectivity, with Ga_2O_3 -doping the best optimization effect. Lu et al. [12] studied La-doping over Cu/ ZrO_2 catalyst and reported that the basic sites accumulate with increasing La loading, resulting in an improved methanol selectivity. Some noble metals such as Pt, Pd, Au, which possess a higher hydrogen dissociation capacity than Cu, are often used in experiments to modify various catalysts [29–31]. These experimental studies have devoted much effort to CO_2 conversion rate and selectivity and less to the influences of doping on the reaction mechanisms.

Focusing on Cu/ZnO-based catalyst, we systematically analyzed the effect of atom-doping on formate pathway to disclose the role of the promoter played in CO_2 hydrogenation. The influence of non-precious metal (Al, Mg, Ga) and noble metal (Pt, Pd, Au) doping on the reaction path was respectively discussed in the levels of the electronic structure, energy and thermodynamic. The mechanisms in the absence and presence of different doping atoms were quantitatively compared based on microkinetic simulation upon DFT calculations, which served as a reliable means to explore the mystery of chemical reactions. The Cu/ZnO-based catalysts was simulated by Cu cluster supported on ZnO surface, which is widely used and believed to better describes the distribution of Cu on ZnO support and the characteristic of the interface [7,16–19]. To gain more insight into the mechanism, a thorough analysis was conducted over the adsorption of several intermediates and the kinetic study over the first CO_2 hydrogenation process was delivered as well. The BEP relationship between the reaction energy and the activation energy of first CO_2 hydrogenation was detected serving as a potential indicator of efficient promoters. The modified database constructed on six element catalysts in this study is expected to provide the theoretical support for the development of the catalysts.

2. Computational details

The DFT calculations are implemented by the VASP 6.1.2 software package [32]. The Projected augmented wave (PAW) method is used to describe the interaction between electrons and ions [33]. Regarding the calculation of the exchange correlation energy, the generalized gradient approximation (GGA) with Poisson-Boltzmann Equation (PBE) is adopted [34]. 400 eV is selected as the plane wave cutoff energy [35]. The integrations of the Brillouin zone were depicted by a $3 \times 3 \times 1$ k-point grid using Monkhorst-Pack Method [32]. Bader charge analysis was conducted to obtain the electron population [37].

Moreover, the transition states (TS) involved in some of the important steps were searched using a constrained minimization scheme with the help of climbing image nudged elastic band (CI-NEB) method as implemented in VASP [36]. The threshold of force convergence for geometry optimization was set less than 0.05 eV [33]. Served as a supplementary, dimer method and the vibrational frequency were calculated to re-confirm the reliability. The TS with merely one imaginary frequency stood out.

The adsorption energy is defined as:

$$E_{\text{ads}} = E_{\text{adsorbate/substrate}} - (E_{\text{adsorbate}} + E_{\text{substrate}}) \quad (4)$$

Herein, E_{ads} represents the adsorption energy of the intermediate in eV; $E_{\text{adsorbate/substrate}}$ indicates the total energy of the entire adsorption system; $E_{\text{adsorbate}}$ implies the energy of the adsorbed molecules in the free state vacuum; $E_{\text{substrate}}$ signifies the energy of the surface system. A positive E_{ads} means that the adsorption is an endothermic reaction, while a negative one indicates that the adsorption is an exothermic reaction. The smaller the value is, the more favorable the adsorption reaction is.

The reaction energy E_r and the activation barrier E_b is defined as:

$$E_r = E_{\text{FS}} - E_{\text{IS}}; E_b = E_{\text{TS}} - E_{\text{IS}} \quad (5)$$

where E_{FS} , E_{IS} , E_{TS} refers to the total energy of the final state, initial state

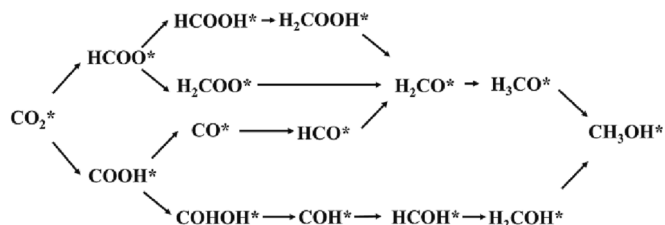


Fig. 1. Reaction path of CO_2 hydrogenation to methanol.

and transition states, respectively.

In this work, a perfect copper crystal with a face-centered cubic structure was constructed (as shown in Fig. 2). Depending on our calculations, the lattice constant is $a = 3.61 \text{ \AA}$, which is in good agreement with the theoretical value of the lattice constant $a = 3.64 \text{ \AA}$ [38]. According to the study of Fishman et al. [39], the surface energies of Cu (111)/Cu(100)/Cu(110) are 1.27/1.44/1.53 $[\text{J/m}^2]$, respectively, indicating that among all three common surface options, Cu(111) surface is the most stable. Müller et al. [40] investigated the adsorption of O^* , H^* , CO_2^* , CO^* and H_2O^* on four crystalline surfaces of Cu(111), Cu(110), Cu(100) and Cu(211) and found that CO_2^* and H_2O^* were both weakly adsorbed on the surfaces; no O^* was adsorbed on any of the crystalline surfaces; and the Cu(111) surface was very dominant in both H^* and CO^* adsorption. They believe that the Cu(111) crystal plane can be regarded as a “hydrogen reservoir” for hydrogenation reactions. Therefore, in this paper, the Cu(111) surface model is selected and a $5 \times 5 \times 2$ Cu(111) surface model is constructed. The unit cell consists of 2 layers of Cu and 50 Cu atoms. A vacuum layer of $10 \text{ \AA}$ vacuum layer was built in the Z direction.

Based on the Cu(111) surface model, a Cu/ZnO catalyst model was constructed, where the relatively stable ZnO(001) surface was selected [41]. The Cu(111) part was constructed above the ZnO(001) to form a heterojunction structure so as to ensure that the mismatch ratio was below 5% [38]. It should be noted that to ensure the continuity at the interface, the lattice parameters of two material layers need to be as compatible as possible, otherwise there is a risk of deformation and collapse of the heterojunction. Only half of the Cu clusters were remained in order to simulate the dispersed arrangement of metal Cu on the carrier ZnO in the actual situation (as shown in Fig. 3a). The first layer (the bottom one) of ZnO is fixed at its equilibrium bulk position, whereas the second layer of ZnO, Cu clusters and adsorption intermediates are relaxed.

The atomic doping modeling was carried out by the single-atom substitution, which is a method commonly used in the modeling of doping. Nunan et al. [42] built a Cs-doped Cu/ZnO model in which a Zn atom was substituted by Cs and found that this modification reduced the activation energy for some of the radical reactions such as HCOOH^* hydrogenation. Similarly, Zhang et al. [43] constructed a single Ni-doped $\text{Ni}_1\text{W}(111)$ surface to conduct the DFT calculations for CO_2 hydrogenation and found that the synergistic effect of W and Ni helps anchor CO_2^* and H^* . Therefore, in this work, a Zn atom (colored yellow) at the interface of a Cu/ZnO-based catalyst model was replaced by the doping atom X to complete the modification (as shown in Fig. 3b).

3. Results and discussion

3.1. Cu/ZnO surface: electronic metal-support interaction (EMSI)

To illustrate the EMSI involved in the Cu/ZnO surface, the details of the Cu/ZnO interface including the electron transfer and the bond length are depicted in Fig. 3c. The distance between Cu and Zn in the interface is $2.580 \text{ \AA}$ and $2.583 \text{ \AA}$, which is shorter than that ($2.684 \text{ \AA}$ [44]) in CuZn, demonstrating the formation of strong interfacial Cu-Zn bonds.

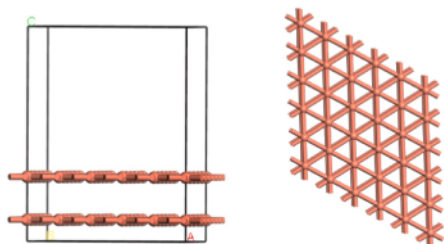


Fig. 2. Surface model of Cu(111) catalyst.

Additionally, the Cu Nanobelt gains 0.52 e from Zn atoms of ZnO, indicating the increased electron density of Cu in Cu/ZnO interface than in pure Cu surface. Actually, we found that for CO_2 adsorption, the Cu sites in Cu/ZnO interface donate more electrons to CO_2 than the Cu sites in the pure Cu(111) surface (0.62 e vs 0.50 e). The adsorbed CO_2 molecule has been better activated (-0.06 eV vs -0.05 eV). These above results demonstrated that there exists the EMSI, in which charge transfer between the Cu and the ZnO support tunes the properties of the site and has enhanced the adsorption. And the EMSI serve as a crucial factor affecting the reactivity for CO_2 hydrogenation. Our calculated results will reinforce the experimental conclusion drawn by Kattel that the interfacial synergy of Cu and ZnO helps facilitate the hydrogenation of CO_2 [18].

The thermodynamic analysis for CH_3OH synthesis on Cu/ZnO and Cu(111) along the formate pathway are shown in Fig. 4, and the geometries of the reaction intermediates are displayed in Fig. 5. In sum, the optimal HCOO^* pathway on Cu/ZnO catalysts follows: $\text{CO}_2^* \rightarrow \text{HCOO}^* \rightarrow \text{HCOOH}^* \rightarrow \text{H}_2\text{COOH}^* \rightarrow \text{H}_2\text{CO}^* \rightarrow \text{H}_3\text{CO}^* \rightarrow \text{CH}_3\text{OH}^*$. Our calculations imply that the presence of ZnO improves the energetic of the intermediates when compared to the case of Cu(111). The reaction occurs at a lower potential energy surface for Cu/ZnO as compared to that for Cu(111). As evidenced in the structure shown in Fig. 5, the HCOO^* pathway starts with the atomic hydrogenation on the C-terminal of CO_2^* . HCOO^* chemisorbs stably at the supporter by the bidentate configuration through two oxygen atoms, which is consistent with the experimental result that HCOO^* could be remarkably detected over the surface [45]. This first hydrogenation step is thermodynamically favorable over both Cu/ZnO and Cu(111) catalyst with a similar adsorption configuration, yet the binding of HCOO^* on Cu/ZnO is 0.36 eV stronger due to the Zn-O reaction (Cu/ZnO: -3.28 eV vs Cu(111): -2.92 eV). Compared to the C-terminal, HCOO^* is preferred to be hydrogenated on O-terminal to produce HCOOH^* over Cu/ZnO rather than CH_2OO^* . This step is an endothermic reaction with a reaction energy of 0.36 eV over Cu/ZnO catalyst, which is 0.15 eV higher than the corresponding energy on the Cu(111) surface. The product H_2COOH^* also anchors through the two oxygen atoms, and the binding of H_2COOH^* is 0.05 eV stronger on Cu/ZnO catalyst. For the dissociation of H_2COOH^* ($\text{H}_2\text{COOH}^* \rightarrow \text{H}_2\text{CO}^* + \text{OH}^*$) on Cu/ZnO catalyst, $E_r = +0.48 \text{ eV}$, which is 0.05 eV lower than that on Cu(111). After the C-O bond breaking, the separated species OH^* and H_2CO^* would be hydrogenated until the production of H_2O^* and CH_3OH^* . The subsequent hydrogenation of H_2CO^* to H_3CO^* ($\text{H}_2\text{CO}^* + 1/2\text{H}_2 \rightarrow \text{H}_3\text{CO}^*$) is facile on Cu/ZnO with E_r as low as -0.23 eV . Compared with that on Cu(111), H_3CO^* on Cu/ZnO is 0.04 eV more stable. Further hydrogenation of H_3CO^* results in the formation of CH_3OH^* ($\text{H}_3\text{CO}^* + 1/2\text{H}_2 \rightarrow \text{CH}_3\text{OH}^*$) ($E_r = -1.10 \text{ eV}$), where CH_3OH^* binds to the Zn site through the O atom. The binding is 0.23 eV stronger than that over Cu(111), and the reaction energy is 0.27 eV lower.

According to the analysis of the potential energy surface in Fig. 4, it is clear that the EMSI between Cu-ZnO will stabilize the O-containing intermediates (HCOO^* , HCOOH^* , H_2COOH^* and H_2CO^*) and decrease the reaction energies for most of the hydrogenation steps. As evidenced by Kattel et al. [8], HCOO^* is an important intermediate in the formate pathway. According to our calculations, Cu/ZnO catalyst is able to significantly increase the adsorption of this important intermediate, which will finally increase the competitiveness of this pathway. As a result, the Cu/ZnO catalyst behave better than Cu(111) surface in terms of the synthesis of methanol from CO_2 hydrogenation in the thermic field. This explains the experimental observations that the Cu/ZnO catalyst resulted in both higher CO_2 conversion and better methanol selectivity than the pure Cu catalyst under the same experimental conditions [24].

3.2. The analysis of doped catalytical surfaces

Our calculations demonstrate that the structure of the Cu/ZnO-X

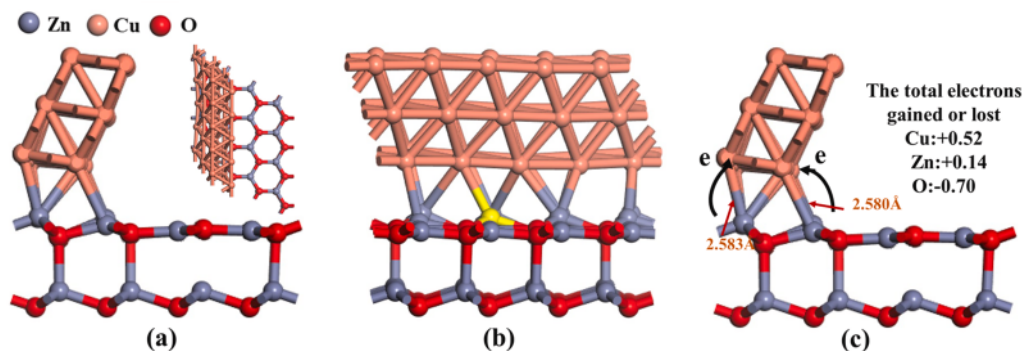


Fig. 3. Model settings for the catalysts. a. Surface model of Cu/ZnO catalyst. b. Model construction of Cu/ZnO-X atomic doping modified catalyst (*Yellow indicates the substituted atom). c. The interfacial bond length and electron transfer of Cu/ZnO catalyst.

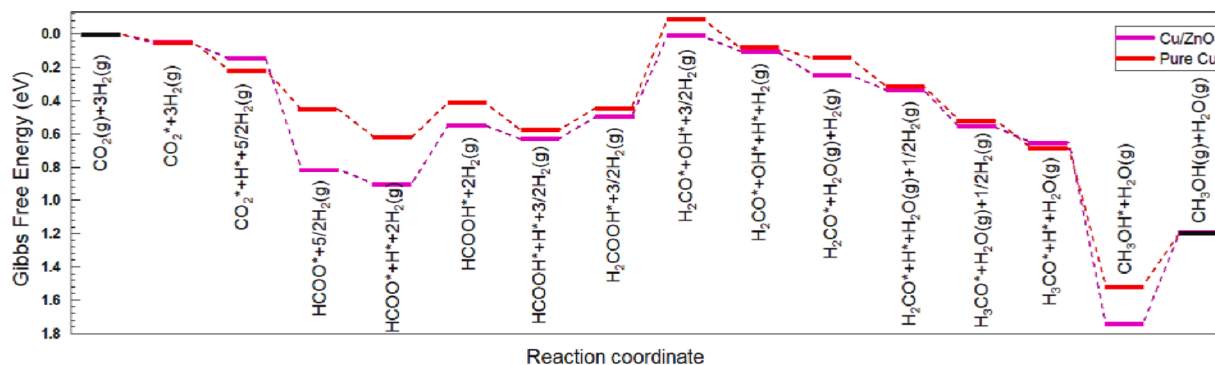


Fig. 4. The thermodynamic analysis for CH_3OH synthesis on Cu/ZnO and Cu(111) along the formate pathway.

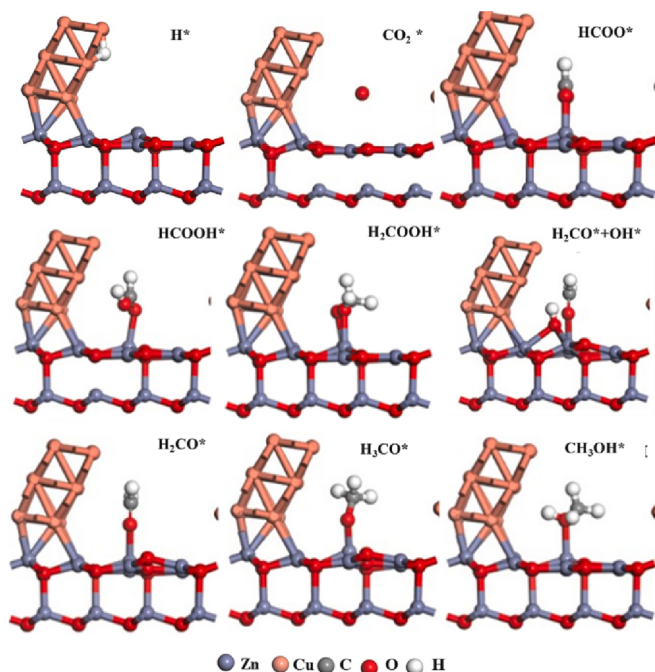


Fig. 5. Structures of the intermediates in the CO_2 methanolization reaction over Cu/ZnO catalyst.

surfaces are almost the same as that of the Cu/ZnO, yet the substituted Cu-X bond length differs. Compared to the optimized results for the non-precious metal doped catalysts, the conformation of the precious metal doped catalysts underwent small changes due to the stronger attraction

between the replaced atom X and Cu cluster. The significant reduction in the bond length of the key bond Cu-X (Cu-Pt, 2.23 Å; Cu-Pd, 2.16 Å; Cu-Au, 1.72 Å; Cu-Ga, 2.43 Å; Cu-Al, 2.59 Å; Cu-Mg, 2.98 Å; Cu-Zn, 2.59 Å, see Fig. S1 in the Supplementary Information(SI)) lead to a relatively poor structural regularity and symmetry, thus a less stable system [46].

To clarify the relative binding properties of the catalysts examined, the d-projected density of states (d-DOSs) together with the d-band center are depicted in Fig. 6. The presented trends correlate well with the increasing d-band center of the surface ϵ_d (unit: eV), defined as [52]:

$$\epsilon_d = \frac{\int \rho E dE}{\int \rho dE} \quad (6)$$

where ρ (unit: states/eV) is the d-DOS and E (unit: eV) is the total energy of the system.

As can be observed in Fig. 6, the wave peaks in the d-band are pronounced both before and after metal atom doping, with a slight shift in the position. The d-band center of Al doping and Pt doping was increased by 0.02 eV, which brings a positive effect on the reduction of the catalytic system. This will be further confirmed in Section 3.3. The study of Norskov et al. [47] revealed that the chemisorption of adsorbates on transition metal surfaces is mainly determined by the bonding between the s and d electronic states of the transition metal surface and the valence electron energy levels of the adsorbate. However, the contribution of the s-states is almost the same for various transition metals, while the d-states varies considerably. Hence, the strength of chemisorption could be depicted by the difference between the d-band energy and the Fermi energy. The higher the d-band center (compared to the Fermi energy), the higher the anti-bond energy, the less the electrons on the anti-key orbital and the stronger the adsorption will be over the post-transition metals.

From the perspective of d-DOS analysis, the doping of Al, Ga, Pt and Pd atoms all enhance the adsorption results, while Mg and Au doping

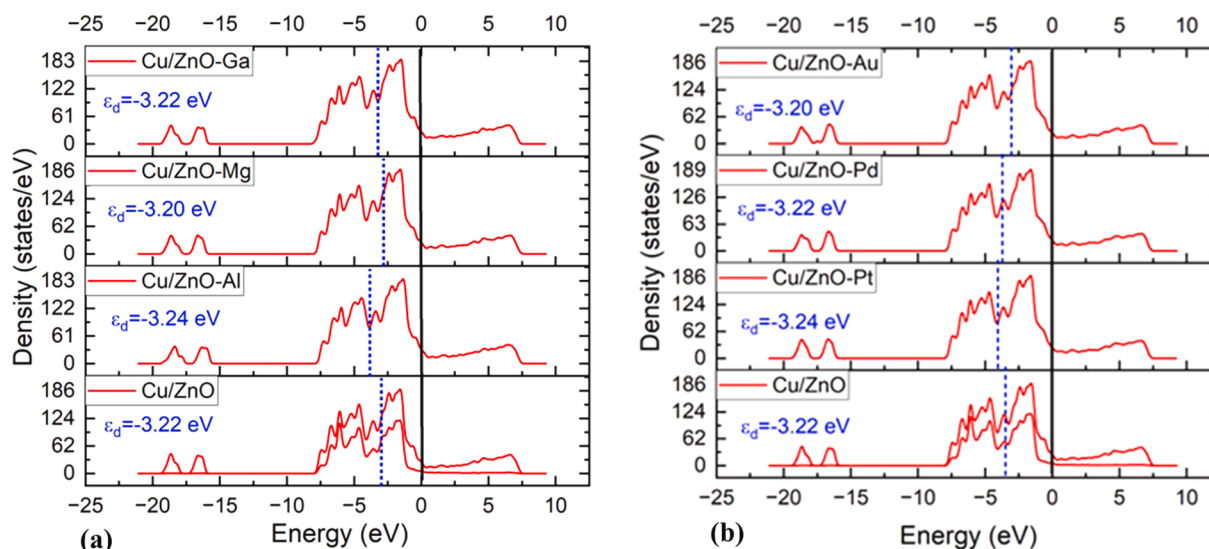


Fig. 6. d-DOSs together with the d-band center calculated for metal doped Cu/ZnO-X, and Cu/ZnO bare surfaces. a. Non-precious metal doped Cu/ZnO-X, and Cu/ZnO bare surfaces. b. Noble metal doped Cu/ZnO-X, and Cu/ZnO bare surfaces.

have a negative impact. However, it is important to note that geometry also has a strong influence on the structure of the adsorbate [48]. Thus, a thorough analysis should be conducted from various perspectives to study the change of the adsorption energy due to the catalyst modification.

3.3. Thermodynamic analysis

According to the experimental and the theoretical results published, there are nine main intermediates involved in the formate pathway [19]. In this part, we will therefore discuss the adsorption of these important

intermediates over the Cu/ZnO-X catalysts. Fig. 7a depicts the thermodynamic analysis for the hydrogenation of CO₂ over Cu/ZnO-X catalysts via the formate pathway. And the adsorption configuration of the intermediates over the catalysts are demonstrated in Fig. S2. The adsorption energies are listed in Table 1. The optimal adsorption sites for most intermediates at the interface of the Cu/ZnO-based catalysts were not altered for each atomic doping modification.

On the one hand, the geometric structure of the intermediates over the doping catalysts of three non-precious metal are very similar to the Cu/ZnO configuration shown in Fig. 4. This is also in agreement with the results obtained experimentally by Schumann et al. [26], further

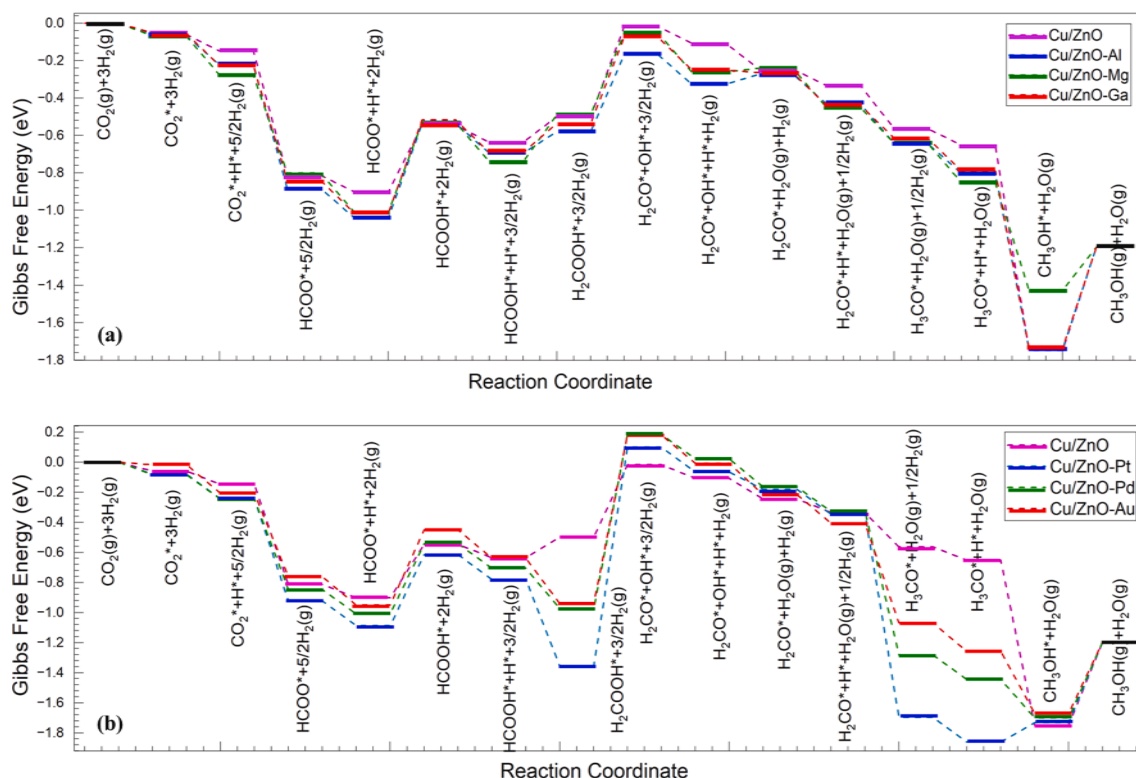


Fig. 7. Thermodynamic analysis for the methanolization of CO₂ over Cu/ZnO based catalysts via the formate pathway. a. Over Cu/ZnO and Cu/ZnO-X (X = Al, Mg, Ga) catalysts. b. Over Cu/ZnO and Cu/ZnO-X (X = Pt, Pd, Au) catalysts.

Table 1

Calculated adsorption energies, E_{ads} (eV) of the intermediates involved in methanol synthesis from CO₂ hydrogenation on Cu/ZnO-X catalysts.

Species	Cu/ ZnO	X = Al	X = Mg	X = Ga	X = Pt	X = Pd	X = Au
H*	-0.09	-0.16	-0.21	-0.17	-0.17	-0.16	-0.19
CO ₂ *	-0.06	-0.06	-0.07	-0.05	-0.08	-0.08	-0.02
HCOO*	-3.28	-3.36	-3.27	-3.31	-3.39	-3.32	-3.23
HCOOH*	-0.37	-0.36	-0.35	-0.34	-0.44	-0.36	-0.27
H ₂ COOH*	-2.45	-2.53	-2.44	-2.48	-3.31	-2.92	-2.89
OH*	-2.21	-2.31	-2.23	-2.21	-2.17	-2.08	-2.09
H ₂ CO*	-0.37	-0.40	-0.36	-0.39	-0.30	-0.29	-0.34
H ₃ CO*	-2.14	-2.22	-2.22	-2.19	-3.28	-2.86	-2.65
CH ₃ OH*	-0.55	0.54	-0.23	-0.54	-0.53	-0.50	-0.47

consolidating the conclusion that the doping with a small amount of non-precious additive will not alter the crystallographic evolution and intermediate adsorption of Cu/ZnO catalyst. Our simulation indicates that the doping of non-precious metals results in more robust H adsorption on the catalyst system, with a doubled adsorption energy over Al and Ga doped catalyst and a tripled (3.56/times) one over Mg doped catalyst (E_{ads} (Clean/Al/Mg/Ga) = -0.09/-0.16/-0.21/-0.17 eV). The activation of CO₂ presents a linear weak physisorption (E_{ads} = -0.06 eV) over all catalysts tested. This weak physisorption will be enhanced slightly with -0.06/-0.07/-0.05 eV (Al/Mg/Ga) respectively. It is worth noting that the energy released during this first hydrogenation (CO₂* + 1/2H → HCOO*) remains the same level (-0.68 ~ -0.52 eV) over all atom-doped catalysts (Table 1). As compared to Cu/ZnO, the change of the reaction energy of the two hydrogenation steps (HCOO* + 1/2H → HCOOH*) and HCOOH* + 1/2H → H₂COOH*) are listed as follows: +0.15/+0.12/+0.14 eV and -0.02/+0.12/+0.02 eV (Al/Mg/Ga resp.). This increment indicates, at least partly, that the contribution of atom doping lies more on the activation of CO₂* rather than the chemisorption of HCOO* and the following hydrogenation steps. The adsorption of HCOOH* performs the worst in this two hydrogenation step (Table 1), which is in line with Wang's conclusion that the poor performance of HCOOH* adsorption restricts the following hydrogenation and finally leads to low methanol yield [19]. After the decomposition of H₂COOH* into H₂CO* and OH*, Al reduces the adsorption energy of H₂CO* by -0.03 eV and thus has a more pronounced effect in stabilizing the adsorption of intermediate structures, which is consistent with the findings of Behren et al. [49].

As can be seen in Fig. 7a, the non-precious metal doping has a thermodynamically significant improvement of CO₂ methanolization via the HCOO pathway over Cu/ZnO based catalysts, where Al > Ga > Mg. The important intermediate HCOO* locates -0.66 eV lower in energy than the initial reactants CO₂(g) and H₂(g) for Cu/ZnO-Al, -0.52 eV lower for Cu/ZnO-Mg, and -0.61 eV lower for Cu/ZnO-Ga. Interestingly, Schumann et al. [21] experimentally prepared Cu/ZnO:M (M = Al, Ga, Mg) catalysts and found that Al and Ga, which have three valence electrons, increase the level of defects in the carrier and thus augment the conductivity of ZnO, while divalent Mg decreases it. As a result, Al and Ga serve as a positive electron promoter in ZnO to reduce the activation energy of some intermediates, while Mg has a more neutral effect on the reaction. Our calculations supplement the current understanding with the finding that Al and Ga with three valence electrons will stabilize some of the important intermediates.

On the other hand, the optimal adsorption configurations of some intermediates (HCOO*, HCOOH*, H₂COOH*) have changed slightly after noble metal doping (X = Pt, Pd, Au) due to the diverse reducibility (see Fig. S2) [30]. Whereas the adsorption of H₂COOH* is one of the key ameliorations in our simulation. In comparison with that over the undoped Cu/ZnO catalyst, the strong attraction of the noble metal to the O atom strengthens the X-O bond and further consolidates the adsorption of H₂COOH*. The intermediate itself deflects more towards the X atom. Consequently, the mean value of the H₂COOH* binding energy

over Cu/ZnO-X (Pt, Pd, Au) is significantly increased by 1.1 eV compared to the case over Cu/ZnO catalyst and Pd-doping takes the top spot with an adsorption energy of -3.31 eV. Furthermore, the noble metal additive shows its advantages in the adsorption of H₃CO*. Shifted from the top site to the hole site, the H₃CO* intermediates over noble metal-doped catalysts holds an average increase in adsorption energy of 0.8 eV (Pt/Pd/Cu: +1.14/+0.72/+0.51 eV respectively).

According to the calculated energetics (Fig. 7b), the noble metal doping has a significant thermodynamic optimization effect on CO₂ methanolization along the HCOO route, where Pt > Pd > Au. As a high quality hydrogenolytic metal, Pd helps mending the hydrogenation process through the hydrogen overflow phenomenon [30]. On the contrary, the intermediates bind more weakly on Cu/ZnO-Au catalyst. The modification of Au-doping neither facilitates the dissociative hydrogenation of H₂COOH* nor stabilize HCOO*. Such results are expected since Au doping enriches the Cu surface with Zn, which is not conducive to the oxidation of Cu [31]. This interesting finding matches well with the experimental results conducted by Lee et al. [50], who doped Ag, Pt, Ru, and Au with a mass fraction of 2% into Cu/ZnO/Al₂O₃ catalyst and found that Pt-doping perform much better than Au-doping in consideration of the conversion of CO₂.

3.4. The linear relationship among the adsorption energy of certain intermediates

Based on the simulation test, Calle-Vallejo et al. [51] found the scaling relationship between adsorption energies of atoms and their hydrides on transition metal surfaces. The adsorption energies of C and S atoms are also linear on transition metal surfaces. Moreover, previous studies also shows that the oxygen-containing intermediates *OOH, *O and *OH all have a somewhat linear relationship in their free energy of adsorption on different catalysts in the field of electrocatalysis [52]. The main reason occurs to be the fact that these oxygen intermediates are all bound to the substrate via an oxygen atom [52]. In this part, we therefore try to investigate the linear relationship between the adsorption energies and of some key intermediates involved in HCOO pathway. The results are shown in Fig. 8 and the detailed information are listed in Table S1.

Two hydrogenation processes show the best relevance (see Figs. 8a and 8b) with the computed $R^2 = 0.78$ for HCOO* + H* → HCOOH* and the computed $R^2 = 0.81$ for H₂CO* + H* → H₃CO*. This kind of linear relationship propose a potential way to predict the reasonable energy range for the adsorption of certain intermediate with hydrogen. But the clear linear relationship of HCOOH* hydrogenation (Fig. 8c) appears only over the noble metal modified catalysts. The strong correlation over the noble metal modified catalysts is probably the results of the strong attraction of the noble metal to the oxygen atoms, which makes the adsorption configuration of the H₂COOH* change compared to HCOOH*, with the original two H atoms deflected more towards the copper clusters (Fig. S2) and thus tighter.

Furthermore, interestingly, the adsorption of H* and CH₃OH* is computed to present a good linear relationship with $R^2 = 0.80$ (Fig. 8c), which highlight the importance of the adsorption of H*. There is a negative correlation between H* adsorption and CH₃OH* adsorption, indicating that the more easily H overflows from a catalyst (with a small adsorption energy of H*), the more strongly CH₃OH* adsorbs and the higher the CH₃OH conversion. These results demonstrate that the heightened hydrogen spillage, the stable and sufficient hydrogen exchange, will help enhance the hydrogenation of CO₂ and increase the conversion rate of methanol. This conclusion is partially in line with Mori's [53] conclusion that hydrogen spillover is the driven factor of CO₂ hydrogenation over CoNiCuRuPd/TiO₂ robust catalyst.

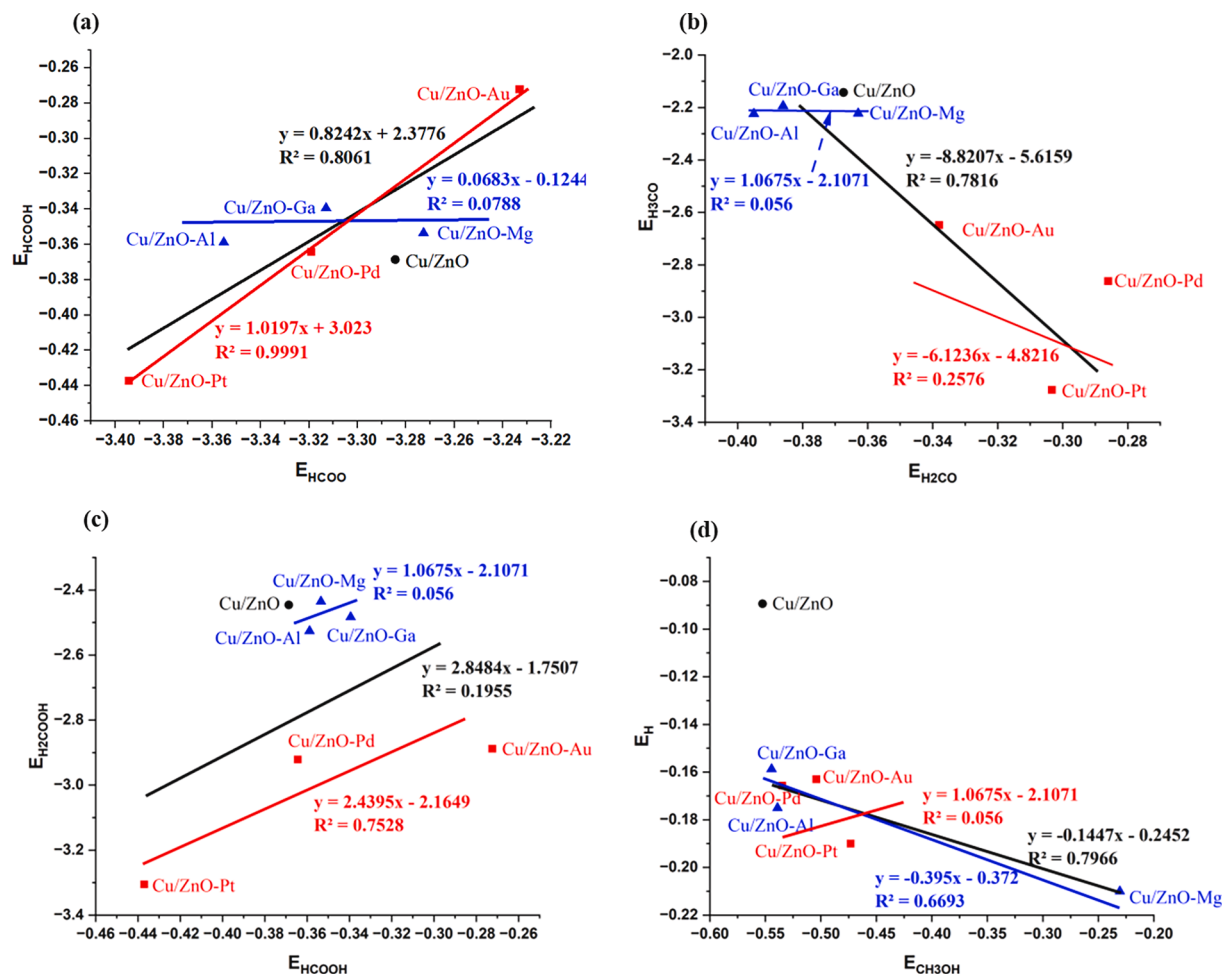


Fig. 8. The linear relationship between the adsorption energy of certain intermediates. a. HCOO^* and HCOOH^* . b. H_2CO^* and H_3CO^* . c. HCOOH^* and H_2COOH^* . d. CH_3OH^* and H^* .

3.5. CO_2 Hydrogenation: The crucial transition state

The hydrogenation of CO_2^* to HCOO^* is of great importance and is served as the rate-limiting step in various CO_2 hydrogenation process towards products like CH_4 , CH_3OH et etc. [54,55]. The main difficulty lies in the relatively high activation barrier of the reaction $\text{CO}_2^* + \text{H}^* \rightarrow \text{HCOO}^*$. Thus, in order to better understand the role of CO_2 activation in

CO_2 hydrogenation process, the transition states are searched and the results are depicted in Fig. 9 and Fig. S4. Al, Ga and Pt doping helps decrease the barrier by 0.07 eV, 0.03 eV, 0.01 eV respectively, while Mg, Pd, Pu increase the barrier by 0.10 eV, 0.03 eV and 0.07 eV. Although half of the doping modification didn't help to reduce the activation barrier E_a , all these modifications manage to low down the highest relative energy over Cu/ZnO (2.56 eV) except Au, the ranking of which

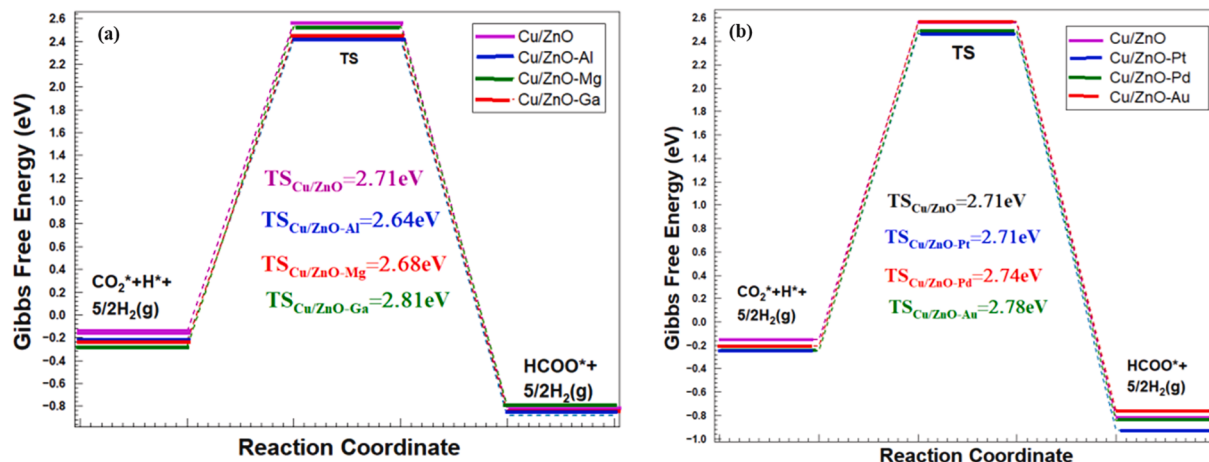


Fig. 9. TS for $\text{CO}_2^* + 1/2\text{H}_2 \rightarrow \text{HCOO}^*$ over Cu/ZnO based catalysts. a. Over Cu/ZnO and Cu/ZnO-X (X = Al, Mg, Ga) catalysts. b. Over Cu/ZnO and Cu/ZnO-X (X = Pt, Pd, Au) catalysts.

follows Cu/ZnO-Al (-0.14 eV) > Cu/ZnO-Ga (-0.11 eV) > Cu/ZnO-Pt (-0.09 eV) > Cu/ZnO-Pd (-0.07 eV) > Cu/ZnO-Mg (-0.03 eV) $\gg$ Cu/ZnO-Au (-0.01 eV).

On the other hand, it has been known that there is a celebrated linear correlation between the activation barrier and the reaction energy of surface reaction on transition metals, which is called the Brønsted – Evans – Polanyi (BEP) relation. Bligaard et al. [56] found the BEP relationship of CO dissociation over transition metal surfaces, drew a theoretical volcano for the production of methane from CO and H₂ basing on the adsorption energy of C and O atom. Therefore, following is focused on the BEP of CO₂ activation in order to simplify the study of the whole complex reaction network into the analysis of adsorption energy. The linearly fitted plot for the hydrogenation of CO₂ to HCOO over all Cu/ZnO-based catalysts is shown in Fig. 10 with the correlation as below:

$E_b(\text{CO}_2) = 0.8449 E_r(\text{CO}_2) - 2.9143$, with a correlation coefficient $R^2 = 0.6519$.

For CO₂ hydrogenation, the barrier increases with the increasing reaction enthalpy. In spite of the fact that the correlation coefficient is hardly perfect, it corresponds well with the BEP relationship of C-H formation from CO₂ over Cu-alloyed catalysts given by Xing et al. [57]. As for the initial state for CO₂ hydrogenation to HCOO*, H* occupies the hollow site and CO₂ weakly interacts with the Cu/ZnO-X surfaces. The adsorption energies of H* and CO₂* are very similar on all Cu/ZnO-X surfaces, indicating that the energy of transition states is strongly influenced by the energy of final states. The adsorption energy of H* turns out to be a potential effective indicator, which is bound to be instructive to further dig in of the whole methanolization process.

4. Conclusion

In the present study, DFT calculations have been used to explore the EMSI effect of Cu/ZnO catalyst and the role of non-precious and noble metal dopants on the catalytic performance of CO₂ hydrogenation to CH₃OH via HCOO pathway on Cu/ZnO-X (X = Al, Mg, Ga, Pt, Pd, Au) catalysts. Most of the intermediates adsorbing through O atom prefer to bind at the Zn site around the interface and they will be strengthened by the doped atoms except for Au. And the variation of the adsorption energies of species on the sites are rather slight, which could be well explained by the minor deviation of the d-band center after single atom-doping. A linear relationship of the hydrogenation of HCOO*, HCOOH* and H₂CO* has been found, indicating a potential way of predicting the data of unknown parameters. The scaling relationship between the adsorption energy of H* and CH₃OH* reinforce the proposal that $E_{a(\text{H}^*)}$ may serve as one of the vital descriptors. The activation barriers of the rate-limiting steps $\text{CO}_2^* + \text{H}^* \rightarrow \text{HCOO}^*$ over diverse modified catalysts are partially decreased, and the whole relative energy decreased. The highest relative energy level follows the rank as: Cu/ZnO-Au > Cu/ZnO-Mg > Cu/ZnO-Pd > Cu/ZnO-Pt > Cu/ZnO-Ga > Cu/ZnO > Cu/ZnO-Al. The calculated results show that the overall reaction process of CH₃OH synthesis is facilitated both kinetically and thermodynamically, especially over Cu/ZnO-Al surface. These findings provide information for the design and optimization of highly efficient Cu/ZnO based catalysts used for CH₃OH formation from CO₂ hydrogenation.

CRedit authorship contribution statement

Xingzi Wang: Conceptualization, Methodology, Software, Investigation, Formal analysis, Visualization, Writing – original draft. **Hai Zhang:** Data curation, Writing – review & editing, Funding acquisition, Resources, Supervision. **Huang Qin:** Conceptualization, Investigation, Software, Resources, Supervision. **Kunming Wu:** Conceptualization, Resources, Supervision. **Kai Wang:** Conceptualization, Validation. **Junfang Ma:** Conceptualization, Resources. **Weidong Fan:** Conceptualization, Funding acquisition, Resources, Supervision.

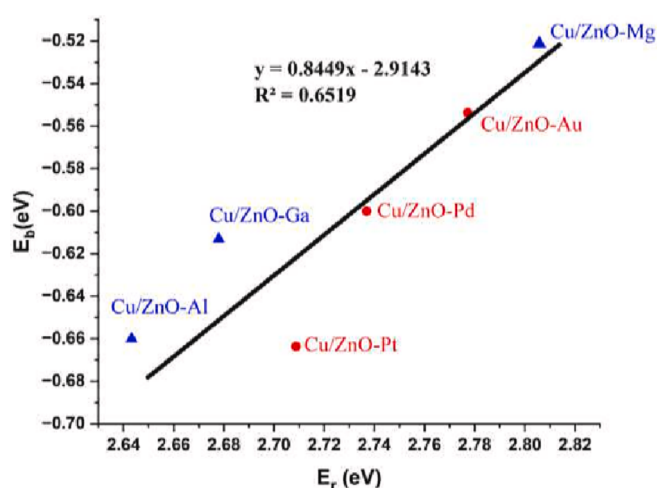


Fig. 10. The BEP relationship of CO₂ hydrogenation over Cu/ZnO-X catalysts (Er refers to the reaction energy and Eb refers to the activation barrier).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fuel.2023.128381>.

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